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REVIEW

How Negative Controls Are Used in Pharmacoepidemiology: A Methodological Scoping Review of Real-World Observational Studies of Glucagon-Like Peptide-1 Receptor Agonists

Zhengxuan Li[1] | Yinan Huang[2,3] | Hua Chen[1] | Jieni Li[1]

[1]Department of Pharmaceutical Health Outcomes and Policy, College of Pharmacy, University of Houston, Houston, Texas, USA | [2]Department of Pharmacy Administration, School of Pharmacy, University of Mississippi, Mississippi, USA | [3]Center for Pharmaceutical Marketing & Management, School of Pharmacy, University of Mississippi, Mississippi, USA

Correspondence: Jieni Li (jli87@central.uh.edu)

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ABSTRACT

Negative controls (NCs) are increasingly used in real-world observational studies to detect residual confounding and systematic bias, but their implementation and reporting remain heterogeneous in applied pharmacoepidemiology. We conducted a methodological scoping review to characterize how NCs were selected, implemented, reported, and interpreted in real-world studies evaluating glucagon-like peptide-1 receptor agonists (GLP-1 RAs) for non-indicated clinical outcomes. We systematically searched PubMed and Embase from database inception through November 2025 and additionally screened reference lists of included studies. Eligible studies used real-world data, applied an observational analytic framework to evaluate GLP-1 RAs, focused on non-indicated outcomes, and explicitly incorporated NCs for bias detection, falsification, validation, or empirical calibration. Among 489 records identified, 42 studies met the inclusion criteria. Most studies were cohort-based and were conducted in diabetes-related populations using electronic health records or administrative claims. Negative control outcomes (NCOs) were the dominant approach, whereas negative control exposures (NCEs), positive controls, and empirical calibration were less common. An explicit rationale for NC selection was reported in most studies, but none of the included studies explicitly discussed the assumptions underlying NC analysis. Most NC findings were reported as null or consistent with expectations, and were primarily used to support validity or robustness rather than to materially alter interpretation. Overall, our findings suggest that the current use of NCs in GLP-1 RA observational research has increased, while standardized implementation and reporting have lagged. Greater transparency regarding why NCs were selected, what source of bias they were intended to probe, how results were interpreted, and whether they changed analytic conclusions may improve the credibility and interpretability of real-world evidence in this rapidly evolving therapeutic area.

1 | Introduction

Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) have expanded rapidly from glucose-lowering therapies for type 2 diabetes to widely used agents across obesity and broader cardiometabolic care [1, 2]. As their use has broadened, so has the volume of real-world evidence (RWE) evaluating GLP-1 RAs across a growing range of non-indicated clinical outcomes,

Key Points

- Negative controls are increasingly used in observational studies of GLP-1 receptor agonists, particularly in recent real-world analyses of non-indicated clinical outcomes.
- In this review of 42 studies, negative control outcomes were far more common than negative control exposures, while positive controls and empirical calibration were used only in a few of studies.
- Most studies reported some rationale for negative control selection, but explicit discussion of the assumptions underlying negative control analysis was absent.
- Negative control findings were usually reported as null or consistent with expectations, and were most often used to support study validity rather than to materially change interpretation.
- More explicit reporting of negative control selection, targeted bias domains, assumptions, quantitative findings, and downstream analytic consequences may strengthen the interpretability of pharmacoepidemiologic real-world evidence.

Plain Language Summary

Studies that use real-world health data are increasingly relied upon to evaluate the benefits and risks of medications, but these studies can be affected by hidden sources of bias. One approach to detecting such bias is the use of "negative controls," which are outcomes or exposures that are not expected to be causally related to the treatment being studied but may still reflect similar sources of confounding or bias. In this review, we examined how negative controls have been used in observational studies of glucagon-like peptide-1 receptor agonists, a widely prescribed class of medications for diabetes and obesity. We found that negative control outcomes were commonly used, but negative control exposures, positive controls, and empirical calibration were much less common. Although many studies briefly explained why negative control was chosen, none explicitly discussed the assumptions required for interpreting negative control findings. In most studies, negative control results were used mainly to support the credibility of the main findings rather than to change conclusions. These results suggest that negative controls are increasingly used in GLP-1 RA research, but clearer and more consistent reporting is needed to improve how readers interpret these analyses.

including cardiovascular, renal, hepatic, neuropsychiatric, ophthalmic, oncologic, gastrointestinal, and other emerging safety or benefit domains [3–5]. This expansion has increased the importance of understanding how bias is assessed in applied observational studies of GLP-1 RAs, particularly when studies address outcomes that were not the original therapeutic targets of the drug class.

RWE studies are particularly valuable in this context because they capture heterogeneous patient populations that are often underrepresented in randomized clinical trials [6, 7]. At the same time, observational evaluations of GLP-1 RAs are vulnerable to several sources of bias [8]. Prescribing decisions are shaped by obesity severity, glycemic status, cardiovascular and renal risk, prior treatment history, healthcare access, and clinician preference, many of which are incompletely measured in administrative claims and electronic health record data [9, 10]. These factors are likely associated with treatment group assignment and can therefore create substantial confounding. Additionally, treatment initiation, intensification, switching, and add-on therapy often occur over time in ways that are associated with both subsequent treatment decisions and outcomes, creating the possibility of time-varying confounding and channeling bias. Differential follow-up intensity, laboratory monitoring, and specialist referral may also contribute to detection or surveillance bias [11]. Even when modern confounding-control approaches are applied, including propensity score methods, high-dimensional propensity scores [12], or marginal structural models [13], residual confounding may persist due to imperfect measurement, unmeasured factors, and heterogeneous care pathways [14–21].

Negative controls (NCs) are increasingly used as an empirical strategy to assess residual bias in observational studies. Negative control outcomes (NCOs) are outcomes that are not expected to be causally affected by the exposure of interest, whereas negative control exposures (NCEs) are exposures that are not expected to causally affect the outcome of interest [22]. In both cases, the key idea is not simply that these controls share measured covariates with the primary exposure-outcome association, but that they may also reflect similar patterns of unmeasured confounding, selection processes, or outcome ascertainment mechanisms [23]. Under formal NC frameworks, this shared confounding structure is often described through the concept of U-comparability, which serves as an alternative to the conditional exchangeability assumption typically invoked in causal inference [23]. As a result, observing expected null associations for appropriately chosen NCs may strengthen confidence in the primary findings, whereas unexpected associations may suggest residual confounding, data quality problems, or violations of analytic assumptions [8, 22–24]. In parallel, positive controls, exposure–outcome pairs with an established causal association and a known direction of effect, are sometimes used alongside NCs to verify that the analytic pipeline can detect a true signal when one is present, thereby complementing the null-association logic of NC analyses.

NC methods have been used in several distinct ways in epidemiologic research. In many applied studies, NCs are used primarily for bias detection or falsification. In large-scale observational analyses, sets of presumed-null outcomes can also be used to characterize systematic error and support empirical calibration of primary effect estimates. Importantly, empirical calibration does not rely on the same formal identification logic as NCO- or NCE-based causal inference approaches, but instead uses the observed distribution of effects across many controls to quantify systematic error. More recently, NC-based methods have also been extended beyond bias detection to contribute directly to causal effect estimation in the presence of unmeasured confounding, including the control outcome calibration approach and double negative control inference frameworks [25–28].

Although methodological frameworks for NC use are now well established [8, 22, 23, 28–30], much less is known about how these tools are used in practice within a single rapidly evolving therapeutic area. A recent broader review [29] summarized the state of NC use in pharmacoepidemiology across diverse settings, but it did not focus specifically on GLP-1 RA studies, which are notable for their rapid growth, broad expansion across indications, and frequent evaluation of non-indicated outcomes. Therefore, we conducted a methodological scoping review of real-world observational studies evaluating non-indicated clinical outcomes of GLP-1 RAs that explicitly incorporated NCs. Our objectives were to characterize how NCs were selected, implemented, and interpreted; to describe the types of NC approaches used, including NCOs, NCEs, positive controls, and empirical calibration; and to examine how NC practices varied across indication groups, study designs, data sources, and outcome domains.

2 | Methods

2.1 | Data Sources and Search Strategy

This methodological scoping review was conducted in accordance with the PRISMA-ScR guidelines [31]. We aimed to identify real-world observational studies evaluating GLP-1 RAs that incorporated NCs for bias assessment, including falsification or bias detection, empirical calibration, and validation. This review focused on the methodological implementation, justification, and reporting of NCs in applied research practice, rather than evaluating the validity of primary causal association estimates reported in individual studies.

We systematically searched PubMed and Embase from database inception through November 2025. Search strategies combined terms for GLP-1 RAs (including individual drug names) and NC methodology (including negative control outcome, negative control exposure, falsification outcome, falsification endpoint, empirical calibration, and bias detection). Full search strategies are provided in Appendix 1. To enhance completeness, we additionally reviewed the reference lists of included studies to identify potentially relevant articles not captured through database searches.

2.2 | Eligibility Criteria and Study Selection

Studies were eligible if they used real-world data sources, applied an observational analytic framework to evaluate GLP-1 RAs, focused on non-indicated clinical outcomes, and explicitly incorporated NCs for falsification or bias detection, empirical calibration, or validation. NCs were considered explicitly incorporated when authors referred to NCs or closely related concepts (e.g., falsification endpoints or calibration outcomes) and applied them to assess bias or interpret findings.

We excluded randomized trials, reviews, editorials, and conference abstracts. Studies that did not incorporate NCs or evaluated only on-label efficacy outcomes were also excluded. Two reviewers independently screened titles and abstracts, followed by full-text reviews. Discrepancies were resolved by consensus. Study identification, screening, eligibility assessment, and inclusion are summarized in Figure 1.

2.3 | Data Extraction and Classification

Two reviewers extracted data using a standardized form. Extracted items included study design, data source, population, exposure and comparator definitions, primary association measures, types of NCs used, implementation of NCs, and key findings.

To address the editorial concern regarding design classification, we abstracted primary design family separately from design features. Retrospective cohort–based studies were grouped under a common cohort-based design family, whereas target

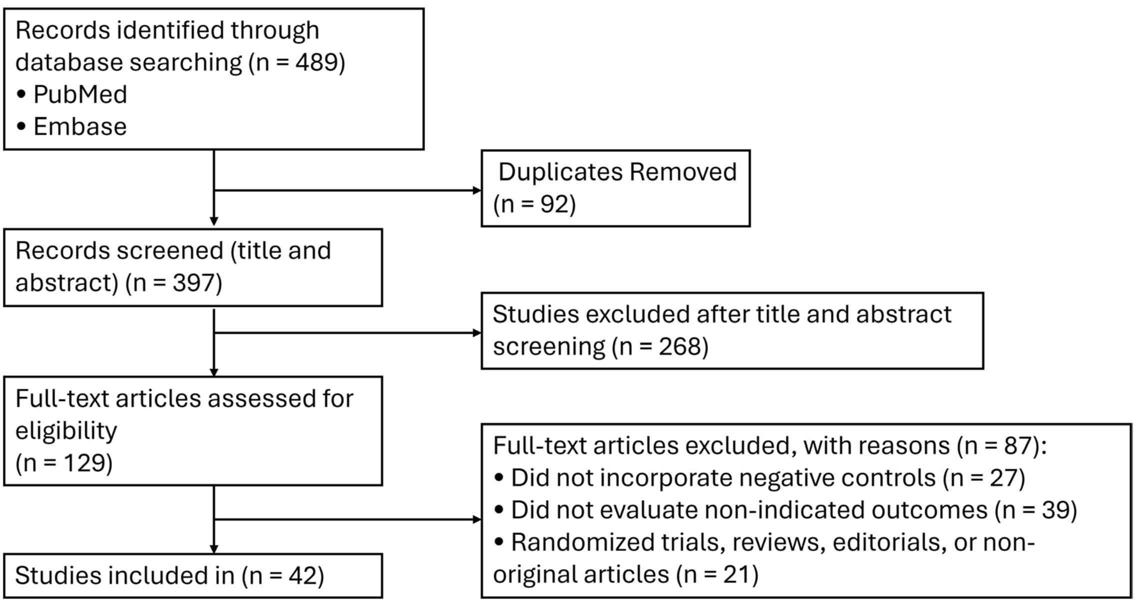

FIGURE 1 | PRISMA flow diagram of study identification, screening, eligibility assessment, and inclusion.

TABLE 1 | Characteristics of included studies.

Characteristic	Category	N	%
Publication year	2021	1	2.4
	2024	13	31.0
	2025	28	66.7
Clinical indication group	Diabetes-related populations	30	71.4
	Obesity-related populations	3	7.1
	Mixed diabetes/ obesity populations	5	11.9
	Other cardiometabolic/ special populations	4	9.5
Primary design family	Cohort-based	38	90.5
	Sequence symmetry analysis	2	4.8
	Case-time-control	1	2.4
	Pharmacovigilance	1	2.4
TTE used	Yes	14	33.3
	No	28	66.7
ACNU used	Yes	23	54.8
	No	19	45.2
Data source family	Electronic health records/ health-system data	25	59.5
	Administrative claims	9	21.4
	Registry/linked databases	6	14.3
	Mixed Electronic health records + claims network	1	2.4
	Pharmacovigilance database	1	2.4

Note: TTE and ACNU were presented as design features rather than primary study design categories.
Abbreviations: ACNU, active-comparator new-user; TTE, target trial emulation.

trial emulation (TTE) and active-comparator new-user (ACNU) designs were coded as separate binary features rather than as mutually exclusive primary study designs. We also abstracted NC-specific items, including control type, number and identity of controls, whether an explicit rationale was reported, the basis of that rationale, whether assumptions were discussed, the reported findings of the NC analysis, and how those findings were handled in study interpretation.

2.4 | Data Synthesis

We summarized included studies descriptively using counts and percentages. Table 1 presents the characteristics of included studies, including publication year, clinical indication

TABLE 2 | Implementation and reporting of negative controls in included studies.

Domain	Category	No. (%) of studies
Type of control analysis	NCO used	38 (90.5)
	NCE used	4 (9.5)
	Both NCO and NCE used	2 (4.8)
	Positive control used	11 (26.2)
	Empirical calibration used	1 (2.4)
	No formal NC framework implemented	1 (2.4)
Rationale and reporting	Explicit rationale for NC selection reported	37 (88.1)
	Assumptions of NC analysis discussed	0 (0.0)
Rationale type	Clinical/subject-matter reasoning	25 (59.5)
	Prior literature/precedent	7 (16.7)
	Study-defined expected null or comparison rationale	4 (9.5)
	OHDSI/empirical calibration framework	1 (2.4)
	No explicit rationale reported	5 (11.9)
Findings of NC analyses	Null/behaved as expected	35 (83.3)
	Mixed or partially non-null findings	4 (9.5)
	Non-null comparator-like signal	1 (2.4)
	Diagnostic/calibration-oriented	1 (2.4)
	No formal NC analysis	1 (2.4)
Handling of NC findings	No material change; used mainly to support validity/robustness	35 (83.3)
	Prompted additional sensitivity analysis/re-analysis	2 (4.8)
	Raised concern but did not overturn main conclusion	2 (4.8)
	Calibration applied	1 (2.4)
	Strengthened main conclusion	1 (2.4)
	No formal NC analysis	1 (2.4)

Note: Categories under "Type of control analysis" are not mutually exclusive. Percentages are based on all included studies ($N = 42$). Rationale types are defined as follows: *clinical/subject-matter reasoning* refers to NCs justified primarily on biological or pharmacological grounds with no plausible mechanism linking them to the exposure; *study-defined expected-null rationale* refers to NCs that the study itself defined as expected-null based on its analytic design rather than on an explicit biological argument. The single study counted under 'No formal NC framework implemented,' 'No formal NC analysis' (Findings), and 'No formal NC analysis' (Handling) is Sarwal et al. 2025, which referenced NC concepts but did not implement a formal NC or positive control analysis; it is retained in the review for completeness.
Abbreviations: NC, negative control; NCE, negative control exposure; NCO, negative control outcome; OHDSI, Observational Health Data Sciences and Informatics.

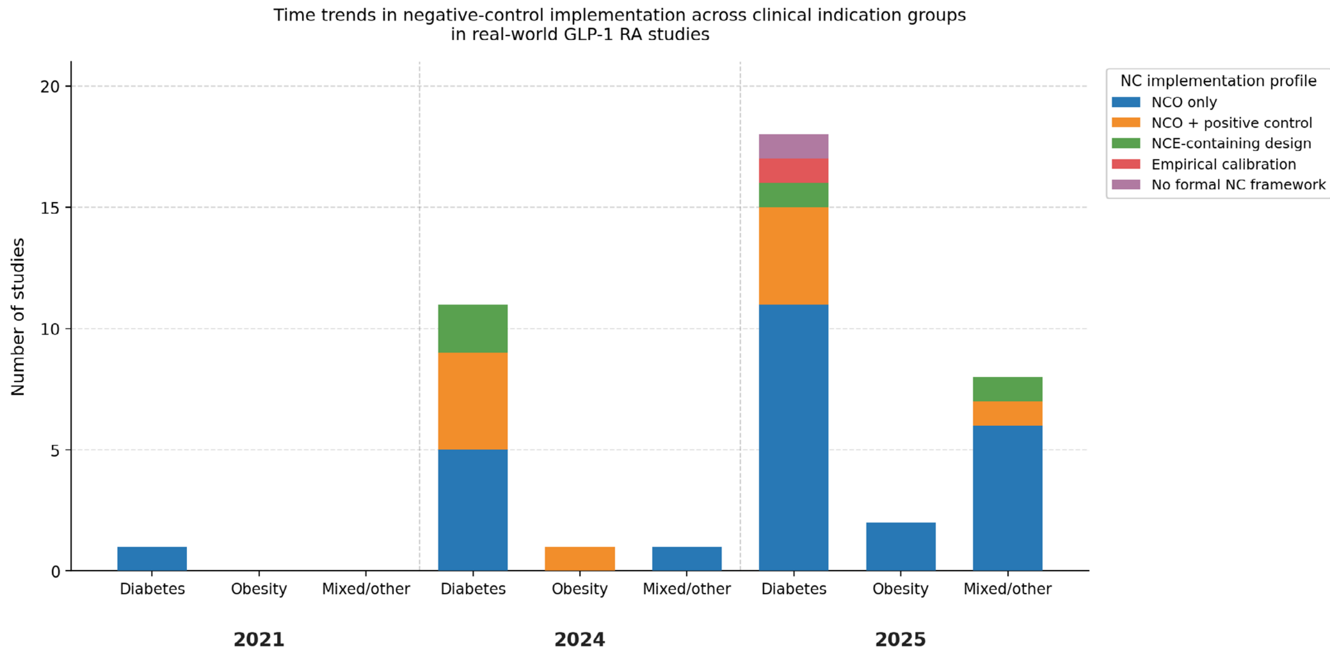

FIGURE 2 | Time trends in negative-control implementation across clinical indication groups in real-world GLP-1 RA studies. Faceted stacked bar charts show the number of included studies by publication year and clinical indication group. Clinical indication groups were classified as diabetes-related populations, obesity-related populations, and mixed/other cardiometabolic populations based on the study population described in the original report. Bar segments represent mutually exclusive negative-control implementation profiles: Negative control outcome (NCO) only, NCO plus positive control, negative control exposure (NCE)-containing design, empirical calibration, and no formal negative-control framework. TTE and ACNU were treated as study design features rather than primary design families. Abbreviations: NC, negative control; NCO, negative control outcome; NCE, negative control exposure.

group, primary design family, use of TTE and ACNU features, and data source family. Table 2 summarizes implementation and reporting of NC analyses, including rationale, whether assumptions were reported, findings, and downstream handling of NC results. Figure 2 presents time trends in NC implementation across clinical indication groups, and Figure 3 summarizes reporting elements, NC findings, and how study authors handled NC results in their interpretation. Because this was a scoping review, we did not perform quantitative synthesis of effect estimates.

3 | Results

3.1 | Study Identification and Selection

The database search identified 489 records from PubMed and Embase. After removal of 92 duplicates, 397 titles and abstracts were screened, and 129 articles underwent full-text review. Of these, 87 were excluded: 27 did not incorporate NCs, 39 did not evaluate non-indicated outcomes, and 21 were randomized trials, reviews, editorials, or otherwise non-original reports. A total of 42 studies [14–21, 32–65] met the inclusion criteria and were included in the final analysis (Figure 1).

3.2 | Study Characteristics

Characteristics of the included studies are summarized in Table 1. Publications were concentrated in recent years, with 28/42 (66.7%) published in 2025 [14, 17–21, 32, 33, 35, 36, 38, 39, 41, 42, 45, 46, 48–53, 57–59, 61, 62, 65], 13/42 (31.0%) in 2024 [15, 16, 34, 40, 43, 44, 47, 54–56, 60, 63, 64], and 1/42 (2.4%) in 2021 [37]. Most studies were conducted in diabetes-related populations (30/42, 71.4%) [15, 16, 18–21, 32, 34–37, 39, 40, 42, 43, 46, 47, 49, 51–54, 56, 57, 59–64], whereas fewer focused on obesity-related populations (3/42, 7.1%) [41, 44, 58], mixed diabetes/obesity populations (5/42, 11.9%) [14, 17, 48, 55, 65], or other cardiometabolic or special populations (e.g., heart failure, chronic kidney disease, or post-transplant cohorts) (4/42, 9.5%) [33, 38, 45, 50]. The included literature was overwhelmingly cohort-based (38/42, 90.5%) [15–21, 32–54, 57–64]. When treated as design features rather than stand-alone primary designs, 14/42 (33.3%) [16, 19, 36, 40, 43, 45, 50–52, 58–61, 64] studies explicitly used a target trial emulation framework and 23/42 (54.8%) [17–21, 33, 34, 36, 37, 39, 42, 43, 45–48, 50, 51, 57, 59–61, 64] used an ACNU design. Data sources were dominated by electronic health records or health-system data (25/42, 59.5%) [15, 17, 20, 21, 32–34, 38–41, 43, 44, 46–50, 54, 57–60, 62, 63], followed by administrative claims (9/42, 21.4%) [16, 35–37, 42, 45, 52, 61, 64], registry or linked databases (6/42, 14.3%) [14, 19, 51, 53, 55, 56], mixed EHR and claims networks (1/42, 2.4%) [18], and pharmacovigilance data (1/42, 2.4%) [65].

3.3 | Patterns of NC Use

Implementation and reporting characteristics of NC analyses are summarized in Table 2. NCOs were the dominant approach, used in 38/42 (90.5%) [15–20, 32–34, 36–55, 57–65] studies. NCEs were used less frequently (4/42, 9.5%) [14, 34, 49, 56], and 2/42 (4.8%) [34, 49] studies incorporated both NCO and NCE elements. Positive controls were used in 11/42 (26.2%) [16, 20, 40, 44, 46, 47, 49, 54, 57, 59, 65] studies, whereas empirical calibration was used in only 1/42 (2.4%)

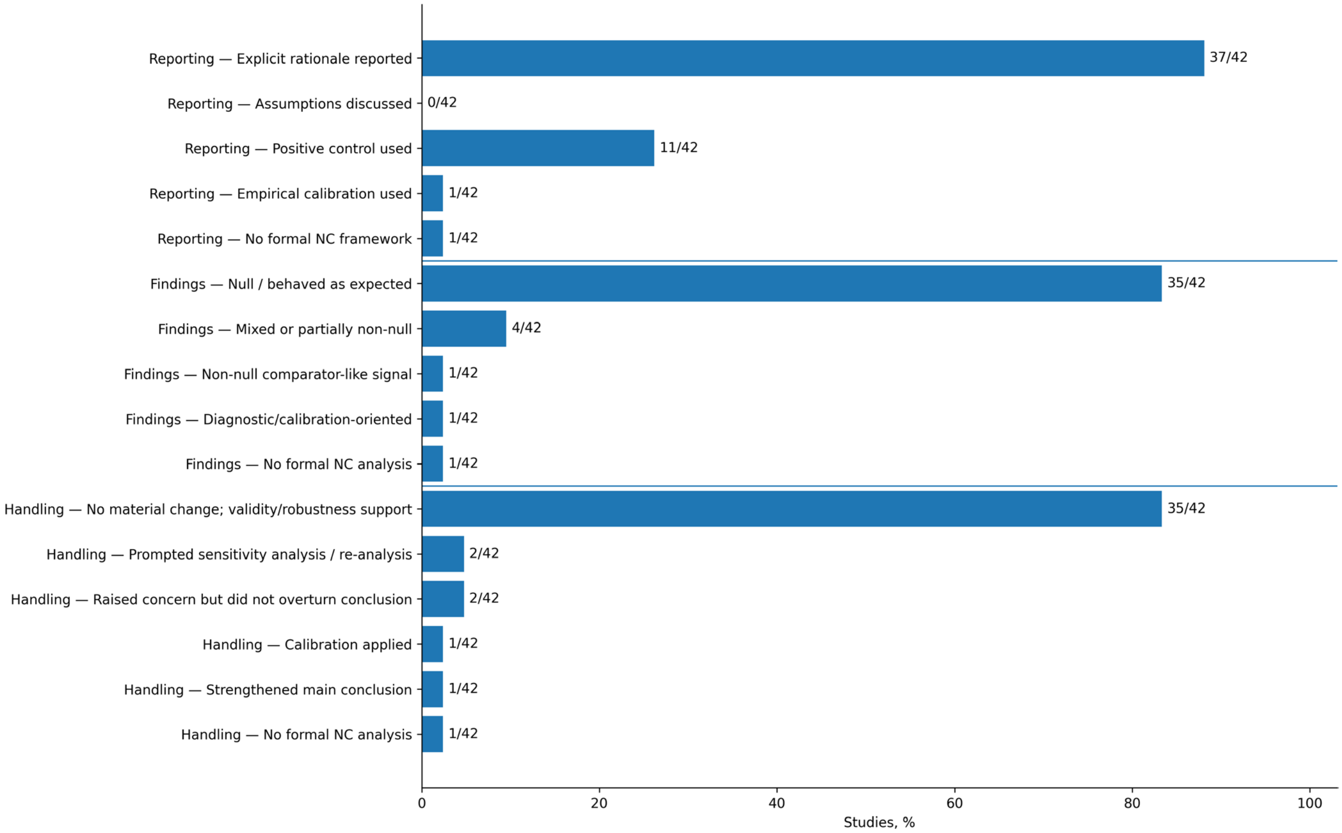

FIGURE 3 | Reporting, findings, and downstream handling of negative-control analyses in included studies. Panel A summarizes reporting elements related to negative-control analyses, including whether an explicit rationale for negative-control selection was reported, whether assumptions were discussed, and whether positive controls or empirical calibration were used. Panel B summarizes the reported findings of negative-control analyses, categorized as null/behaved as expected, mixed or partially non-null, non-null comparator-like signal, diagnostic/calibration-oriented, or no formal negative-control analysis. Panel C summarizes how study authors handled negative-control findings in the interpretation of their main analyses, including whether findings were used primarily to support validity, prompted additional sensitivity analyses or re-analysis, raised concern without overturning the main conclusion, or strengthened the main conclusion. Percentages are based on all included studies ($N = 42$). Abbreviations: NC, negative control.

[35] study. One included study referenced an NC framework conceptually but did not implement a formal NC analysis [21]; specifically, Sarwal et al. relied on an active-comparator new-user design with three comparator arms and used the expected mortality difference between GLP-1 RA and insulin glargine as an implicit pipeline check, rather than implementing a prespecified NC or positive control. This study is retained in our review because the authors explicitly invoked NC reasoning but is classified throughout Table 2 as 'No formal NC framework implemented'. Time trends in NC implementation are shown in Figure 2. NC use in this literature was concentrated in 2024–2025 and was most common in diabetes-related populations. Across indication groups, NCO-only strategies predominated, while positive controls, NCE-containing designs, and empirical calibration were much less common.

3.4 | Rationale, Assumptions, Findings, and Downstream Handling of NC Analyses

An explicit rationale for NC selection was reported in 37/42 (88.1%) [14–16, 18–20, 32–35, 37–47, 49–59, 61–65] studies. The most common rationale type was clinical or subject-matter reasoning (25/42, 59.5%) [14, 19, 20, 32, 34, 39–47, 49–53, 55–57, 62, 63, 65], followed by prior literature or methodological precedent (7/42, 16.7%) [15, 16, 54, 58, 59, 61, 64] and study-defined expected-null or comparison rationale (4/42, 9.5%) [18, 33, 37, 38]. Where a targeted bias domain could be inferred from the published rationale, residual or unmeasured confounding in general was the most frequently cited concern (30/42, 71.4%) [14, 16–18, 20, 21, 33, 35–40, 42–48, 50, 52–54, 57, 59–61, 63, 64], followed by differential healthcare utilization, health-seeking behavior, or healthy-user bias (13/42, 31.0%) [14, 15, 19–21, 35, 37, 42, 43, 55, 57, 58, 62]; selection bias (5/42, 11.9%) [38, 46, 47, 49, 53]; reporting, protopathic, survival, or other timing-related biases (5/42, 11.9%) [14, 19, 50, 55, 65]; detection or surveillance bias, including differential outcome ascertainment (4/42, 9.5%) [20, 33, 40, 51]; and confounding by indication (3/42, 7.1%) [16, 56, 65]. Five studies (11.9%) [32, 41, 49, 56, 62] provided a rationale but did not specify the unmeasured confounder or bias source the control was intended to interrogate. Categories are not mutually exclusive, as several studies invoked more than one bias domain. Only 1/42 (2.4%) [35] study used a broader OHDSI/empirical calibration framework as the main rationale for NC selection, and 5/42 (11.9%) [17, 21, 36, 48, 60] did not report an explicit rationale. Despite the relatively frequent reporting of selection rationale, none of the included studies explicitly discussed the assumptions of NC analysis in the reported results (0/42, 0.0%). Most NC analyses were reported as null or as behaving as expected (35/42, 83.3%) [15–17, 20, 32–41, 43–54, 56–64]. In most studies, NC findings did not materially change the main study conclusion and were used primarily to support validity or robustness (35/42, 83.3%) [15–17, 20, 32–41, 43–54, 56–64], whereas a minority prompted additional sensitivity analyses or re-analysis (2/42, 4.8%) [42, 65], introduced caution without overturning conclusions (2/42, 4.8%) [19, 55], or were incorporated into a calibration framework (1/42, 2.4%) [35] (Figure 3).

Detailed study-level characteristics and NC findings are provided in Table S1 (Part A and B).

4 | Discussion

4.1 | Principal Findings

In this methodological scoping review of 42 real-world observational studies of GLP-1 RAs evaluating non-indicated clinical outcomes, we found that NC use was increasingly common but methodologically heterogeneous. The literature was concentrated in 2024–2025, dominated by cohort-based designs, and largely conducted in diabetes-related populations using electronic health records or administrative claims. NCOs were the predominant approach, used in 38 of 42 studies (90.5%), whereas NCEs were incorporated in only 4 (9.5%), positive controls in 11 (26.2%), and empirical calibration in only 1 (2.4%). An explicit rationale for NC selection was reported in 37 studies (88.1%), but no study explicitly discussed the assumptions underlying the NC analysis (0/42, 0.0%). NC findings were null or described as consistent with expectations in 35 studies (83.3%); mixed or partially non-null findings were observed in 4 (9.5%). In most studies, NC results did not materially alter the primary interpretation: only 2 studies (4.8%) used NC findings to trigger additional sensitivity analyses, 2 (4.8%) introduced caution without overturning conclusions, and 1 (2.4%) applied empirical calibration.

These findings suggest that NCs have become part of the applied analytic vocabulary of contemporary GLP-1 RA RWE, but their use is not yet standardized. In practice, NCs were most often implemented as pragmatic falsification tools rather than as part of a more explicit causal framework. This distinction matters. The presence of an NC analysis does not by itself indicate that the assumptions underlying the chosen control are clear, that the control meaningfully captures the main source of concern, or that unexpected NC results would actually modify study interpretation. Our review therefore suggests that the current literature has moved further in the direction of including NCs than in the direction of fully specifying how and why they should be interpreted.

4.2 | Rationale, Assumptions, and What Was Actually Reported

A major finding of this review was the disconnect between the frequency of NC use and the limited depth of reporting surrounding their rationale. Most included studies gave at least some justification for NC selection, but those rationales were commonly brief and qualitative. The most common basis was broad clinical or biological implausibility (25/42, 59.5%), followed by reference to prior literature or methodological precedent (7/42, 16.7%), a study-defined expected-null comparison (4/42, 9.5%), and an OHDSI-based empirical calibration framework (1/42, 2.4%). Five studies (11.9%) reported no explicit rationale at all. Across all included studies, none explicitly discussed the assumptions underlying their NC approach in the published report (0/42, 0.0%).

This gap is important because the usefulness of an NC depends not only on presumed causal independence from the primary exposure or outcome, but also on whether the control reflects similar sources of unmeasured confounding, selection bias, or outcome ascertainment as the primary exposure-outcome association [8, 11, 22–24]. In formal NC frameworks, this property is often described through the concept of U-comparability, the requirement that the NC shares the same unmeasured confounding structure as the primary analysis, which serves as a substitute for the conditional exchangeability assumption typically invoked in causal inference [23]. In principle, an NCO chosen to reflect frailty, healthcare utilization, access to care, or surveillance intensity provides more interpretable information than an outcome selected solely because it appears clinically unrelated to the drug. Likewise, NCEs are most informative when they plausibly share the confounding structure of the primary exposure without causally affecting the outcome of interest [22]. In our review, however, these considerations were largely implicit. No included study explicitly stated which source of unmeasured confounding its NC was intended to capture, nor whether the chosen control was expected to satisfy the U-comparability condition. That absence limits interpretability: if the underlying assumptions are not stated, null NC findings may be treated as broadly reassuring even when it is unclear what specific bias they were intended to interrogate, and non-null findings may be difficult to act on beyond a generic acknowledgment that residual bias may remain.

4.3 | Findings of NC Analyses and Their Downstream Use

Another important observation was that most NC findings were reported as null or as behaving as expected, and most did not materially alter the interpretation of the primary study. In many articles, NC analyses functioned as supportive robustness checks appended to the main analysis rather than as decision points that could redirect interpretation. This does not make the analyses uninformative. On the contrary, null NC findings can provide meaningful reassurance when they are well matched to the suspected bias structure of the primary analysis and when the assumptions justifying that match are transparent. However, our results indicate that in the current GLP-1 RA literature, NCs were more often used to support credibility than to challenge conclusions or trigger reanalysis.

Only 4 studies (9.5%) [19, 42, 55, 65] reported mixed or partially non-null NC findings. Of these, two reported a borderline non-null signal that was explicitly acknowledged but did not overturn the primary conclusion [19, 55], one reported mixed NC results in the main analysis that resolved to null in a sensitivity analysis [65], and one observed a significant NCO signal in one of two analytic databases that prompted mandatory additional analyses using alternative weighting approaches, ultimately yielding consistent results [42]. Notably, one study using an NCE design [14], in which DPP-4 inhibitors served as the negative control exposure, observed a similar inverse association between the NCE and suicidal behavior as was observed for GLP-1 RAs, leading the authors to conclude that the apparent protective effect was more likely attributable to reduced healthcare utilization before suicidal acts than to a true pharmacological effect of GLP-1 RAs. This example illustrates both the interpretive value of NCEs when they behave unexpectedly and the importance of prespecifying what action will be taken if an NC analysis yields a non-null result.

Empirical calibration, which uses a large set of presumed-null outcomes to characterize and adjust for systematic error in the observational data, rather than relying on formal causal identification conditions, was used in only 1 of 42 studies (2.4%) [35]. In that study, the well-centered empirical null distribution across approximately 60 NCOs indicated that residual systematic error was minimal and that calibration had little impact on the primary estimates, lending additional credibility to the main findings. The rarity of empirical calibration in the remaining literature suggests that this approach, despite its potential to quantify rather than simply detect systematic bias, has not yet been widely adopted in applied GLP-1 RA research.

4.4 | Comparison With Prior Methodological Reviews

Our findings are broadly consistent with, but extend beyond, two prior reviews of NC use in epidemiology and pharmacoepidemiology. Yang and colleagues [29] conducted a scoping review of 37 methodological articles on NCs published through 2022, mapping the development of NC methods along three axes: bias detection, *p*-value calibration, and causal effect correction using double negative control and proximal inference frameworks. They concluded that methodological work on NCs has advanced rapidly, but that practical guidance on how applied researchers should select NCs, verify their underlying assumptions, and interpret non-null NC results remained limited. Zafari and colleagues [8] complemented that methodological perspective with an applied review of 184 pharmacoepidemiologic studies using NCs through September 2020, in which NCOs were the most common approach (used in approximately 50% of included studies), bias detection was the dominant application (81%), and assumption checking was rarely reported.

Our GLP-1 RA-focused review differs from both in scope and complements them in findings. Whereas Yang et al. cataloged the space of NC methods as described in methodological literature, our review characterizes how those methods are actually implemented in applied real-world studies within a single rapidly evolving therapeutic area. Whereas Zafari et al. synthesized applied practice across all of pharmacoepidemiology, our review restricts attention to GLP-1 RA studies of non-indicated outcomes, where concerns about residual confounding, channeling, time-varying treatment, and detection bias are especially salient. Against this backdrop, three features of the GLP-1 RA literature stand out. First, the proportion of studies using NCOs in our sample (90.5%) was substantially higher than the approximately 50% reported by Zafari et al., suggesting that NCO-based falsification has become nearly universal in this therapeutic area, likely reflecting both the rapid expansion of GLP-1 RA outcome research and the influence of large EHR-network platforms such as TriNetX that make NCO implementation operationally straightforward. Second, consistent with both prior reviews, we found that assumptions reporting was essentially absent (0/42, 0.0%) even in recent studies using sophisticated design features such as target trial emulation and ACNU frameworks, underscoring that this is a persistent gap across pharmacoepidemiology rather than a problem unique to GLP-1 RA research, and one that the methodological advances cataloged by Yang et al. have not yet penetrated applied practice. Third, whereas Zafari et al. observed that bias correction and *p*-value calibration together accounted for approximately 13% of NC utility in their sample, empirical calibration was used in only 1 of 42 studies (2.4%) [35] in our GLP-1 RA sample, and none of the included studies applied the causal-effect-correction methods (e.g., control outcome calibration, double negative control inference) that were central to the methodological landscape described by Yang et al. Taken together, these comparisons indicate that the GLP-1 RA literature reflects broader patterns in pharmacoepidemiology, but with a more pronounced dominance of NCO-only falsification, lower uptake of assumption transparency and formal calibration methods, and essentially no penetration of more advanced NC-based causal inference approaches.

4.5 | Implications for Reporting and Operational Guidance

One practical implication of this review is that the field would benefit from more explicit reporting standards for applied NC analyses. At minimum, studies using NCs should clearly report: (1) whether the control is an NCO, NCE, positive control, or part of an empirical calibration framework; (2) the specific source of bias the control is intended to probe (e.g., frailty, healthcare-seeking behavior, differential monitoring, or residual confounding by disease severity), and the basis for that selection (prespecified, literature-based, clinically motivated, or algorithmically derived); (3) whether the chosen control is expected to share the unmeasured confounding structure of the primary analysis and how that expectation is justified; (4) the quantitative results of the NC analysis; and (5) how those findings affected interpretation, including whether they prompted reanalysis, added caution, supported calibration, or were judged not to materially affect the main conclusion. For studies using empirical calibration, authors should additionally describe how the control set was assembled, how many controls were included, and whether calibrated estimates differed meaningfully from uncalibrated ones.

Such reporting would help translate theoretical NC considerations into operational elements that readers can evaluate and would reduce the tendency for NCs to be presented as generic validity checks without a clear link to the specific bias structure of the study. Beyond falsification and empirical calibration, it is also worth noting that NC-based methods have been extended to support direct causal effect estimation in the presence of unmeasured confounding, including the control outcome calibration approach and double negative control inference frameworks [25, 26, 66]. None of the included studies in our review applied these more advanced methods, suggesting that their adoption in applied GLP-1 RA research remains a future opportunity, particularly for outcomes where unmeasured confounding is substantial and difficult to address through design alone.

4.6 | Strengths and Limitations

This review has several strengths. It focuses on a rapidly growing body of GLP-1 RA literature in which concerns about residual confounding, channeling, time-varying treatment patterns, and detection bias are especially salient. It synthesizes current applied practice across diverse data sources, including EHR networks, administrative claims databases, national registries, linked registry-pharmacy databases, and pharmacovigilance systems, providing a broad picture of how NC methods are used in practice rather than how they are described in methodological guidance. The structured abstraction framework also allowed us to move beyond simple counts of whether NCs were used and instead characterize rationale, assumptions reporting, findings, and downstream handling in a systematic way.

Several limitations should also be considered. First, this was a descriptive scoping review; we did not evaluate the empirical validity of specific NC choices or determine whether a given control truly satisfied the causal-independence and U-comparability assumptions required for valid inference. Second, our synthesis depended on what authors explicitly reported, and some studies may have considered assumptions or rationale more carefully than was described in the published article. Third, some classification decisions, such as grouping indication areas or categorizing how NC findings were handled, required structured abstraction of heterogeneous study narratives, and reasonable analysts might classify individual studies differently. Fourth, because our review focused specifically on GLP-1 RA studies of non-indicated outcomes, the findings may not fully generalize to other rapidly evolving therapeutic areas. Finally, although this focused scope is a strength for characterizing applied practice within a single drug class, it also means that our conclusions describe current reporting and implementation patterns rather than adjudicating the validity of any specific substantive clinical association.

5 | Conclusions

Real-world observational studies of GLP-1 RAs have increasingly incorporated NCs as part of their analytic validation strategy, but the current literature reflects a gap between the prevalence of NC use and the depth of methodological transparency surrounding it. NCOs dominated the applied landscape; assumptions were rarely if ever articulated, and NC findings were overwhelmingly treated as supportive rather than as active decision points. NCEs, positive controls, and empirical calibration, approaches that can provide more structured and actionable information about the nature and magnitude of systematic bias, remained underutilized. As the GLP-1 RA evidence base continues to expand into increasingly diverse and off-label outcome domains, clearer reporting of NC rationale, assumptions, and downstream handling will be essential for readers to evaluate whether observed associations reflect true drug effects or residual bias. The field would benefit from operational reporting guidance that translates the theoretical logic of NC methods into consistent, checkable elements within published studies.

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Ethics Statement

This study was based on analysis of previously published literature and did not involve human participants. Institutional review board approval and informed consent were not required.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data underlying this study are derived from published articles identified through the literature search. The detailed search strategies, study selection process, and extracted study-level information are provided in the Supporting Informations. No original individual-level data were generated for this study.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Characteristics of real-world studies of GLP-1 receptor agonists using negative controls. **Table S1A:** General study characteristics—Part 1. **Table S1A:** General study characteristics—Part 2. **Table S1B:** Negative control details—Part 1. **Table S1B:** Negative control details—Part 2. **Table S1B:** Negative control details—Part 3. **Appendix 1:** Search strategies and keyword combinations.